National University of Computer & Emerging Sciences Islamabad

FAST School of Computing **Fall 2026** **Islamabad Campus**

MT1004 – Linear Algebra
Homework # 3

Part 1: Data Center Network Flow (Non-Homogeneous Systems)

Context: In computer networking, the flow of data packets through server nodes must be strictly conserved (Data In = Data Out). We can model a network of servers as a system of linear equations to identify bottlenecks and redundant pathways.

Consider a localized server cluster with 4 routing nodes (A, B, C, D) and 5 internal data pathways $(x_1, x_2, x_3, x_4, x_5)$ measured in terabytes per second (TB/s).

- **Node A:** Receives 500 TB/s from the external internet. It routes this data out through pathways x_1 and x_2.
- **Node B:** Receives data from x_1 and another external source providing 200 TB/s. It routes data out through pathways x_3 and x_4.
- **Node C:** Receives data from x_2 and x_3. It routes data out to the external internet at a rate of 400 TB/s, and sends the rest through pathway x_5.
- **Node D:** Receives data from x_4 and x_5. It routes all of this to external storage at a rate of 300 TB/s.

Questions:

1. Write down the linear equation for each node based on the principle of flow conservation (Total Flow In = Total Flow Out).
2. Rearrange the system so all variables $x_1 \dots x_5$ are on the left, and set up the augmented matrix.
3. Find the general solution of the network flow by hand, expressing it in parametric vector form.
4. What does the particular solution ($\vec{p}$) represent in physical terms? What do the homogeneous vectors ($\vec{u}$ and $\vec{v}$) represent in terms of data circulation within the network?

Part 2: Audio Signal Compression (Linear Independence)

Context: In digital signal processing and data compression algorithms, sending redundant data wastes bandwidth. If a digital signal can be represented as a linear combination of other signals, it is linearly dependent and can be compressed or dropped.

You are designing a compression algorithm for a music streaming service. You sample 4 audio waveforms over 4 milliseconds. Each signal is stored as a vector in $\mathbb{R}^4$:

$$\vec{s}_1 = \begin{bmatrix}1\\2\\1\\1\end{bmatrix}, \quad \vec{s}_2 = \begin{bmatrix}2\\4\\0\\-2\end{bmatrix}, \quad \vec{s}_3 = \begin{bmatrix}4\\8\\2\\0\end{bmatrix}, \quad \vec{s}_4 = \begin{bmatrix}0\\0\\2\\4\end{bmatrix}$$

Let the signal matrix be $S = \begin{bmatrix}\vec{s}_1 & \vec{s}_2 & \vec{s}_3 & \vec{s}_4\end{bmatrix}$.

Questions:

1. Write the homogeneous matrix equation $S\vec{x} = \vec{0}$ as an augmented matrix and row-reduce it to find the pivot columns.
2. Show that the four audio signals are not linearly independent. Justify your answer using the definition of linear independence.
3. Find a non-trivial linear dependence relation among the signals (e.g., $c_1\vec{s}_1 + c_2\vec{s}_2 + c_3\vec{s}_3 + c_4\vec{s}_4 = \vec{0}$).
4. To save bandwidth, your algorithm should only transmit a subset of independent signals. Which signals act as the "basic" signals (basis), and how can the redundant signals be reconstructed strictly from the basic ones?

Part 3: Python Implementation (Coding Deliverable)

Write a Python script to computationally verify the math you solved by hand.

Task 1 (Network Flow):

Use `sympy.Matrix` to define the augmented matrix from Part 1. Use the `.rref()` method to compute the reduced row echelon form. Simply print the resulting RREF matrix and the tuple of pivot columns that `sympy` returns. Add a brief Python comment explaining how you can visually verify your hand-written parametric vector form from this output.

Task 2 (Signal Independence):

Using `sympy`, define the augmented matrix for the homogeneous system $[S \mid \vec{0}]$ from Part 2.

- Use the `.rref()` method to find the reduced row echelon form.
- Write logic in your code to check the number of pivot columns versus the number of variables.
- Write a print statement that outputs: *"The signals are linearly dependent because the system has free variables, meaning the homogeneous equation $S\vec{x} = \vec{0}$ has non-trivial solutions."*